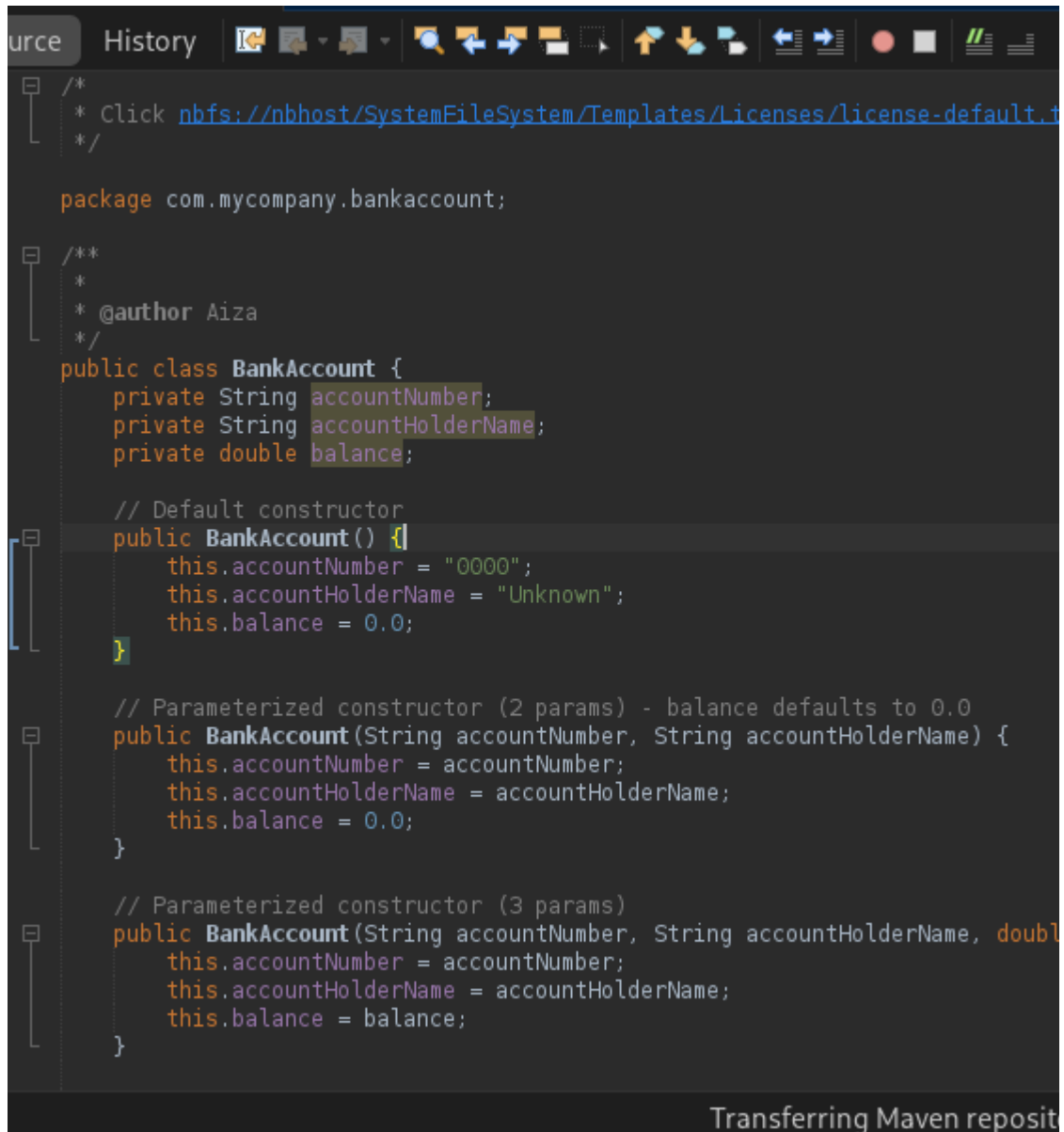


OOP LAB #3

Task#1:



The screenshot shows a code editor with the following Java code:

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.t
 */

package com.mycompany.bankaccount;

/**
 *
 * @author Aiza
 */
public class BankAccount {
    private String accountNumber;
    private String accountHolderName;
    private double balance;

    // Default constructor
    public BankAccount() {
        this.accountNumber = "0000";
        this.accountHolderName = "Unknown";
        this.balance = 0.0;
    }

    // Parameterized constructor (2 params) - balance defaults to 0.0
    public BankAccount(String accountNumber, String accountHolderName) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = 0.0;
    }

    // Parameterized constructor (3 params)
    public BankAccount(String accountNumber, String accountHolderName, double balance) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = balance;
    }
}
```

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OutPut:

```
--- resources:3.3.1:resources (default-resources) @ BankAccount ---
skip non existing resourceDirectory /home/rehan/NetBeansProj

--- compiler:3.13.0:compile (default-compile) @ BankAccount ---
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to target

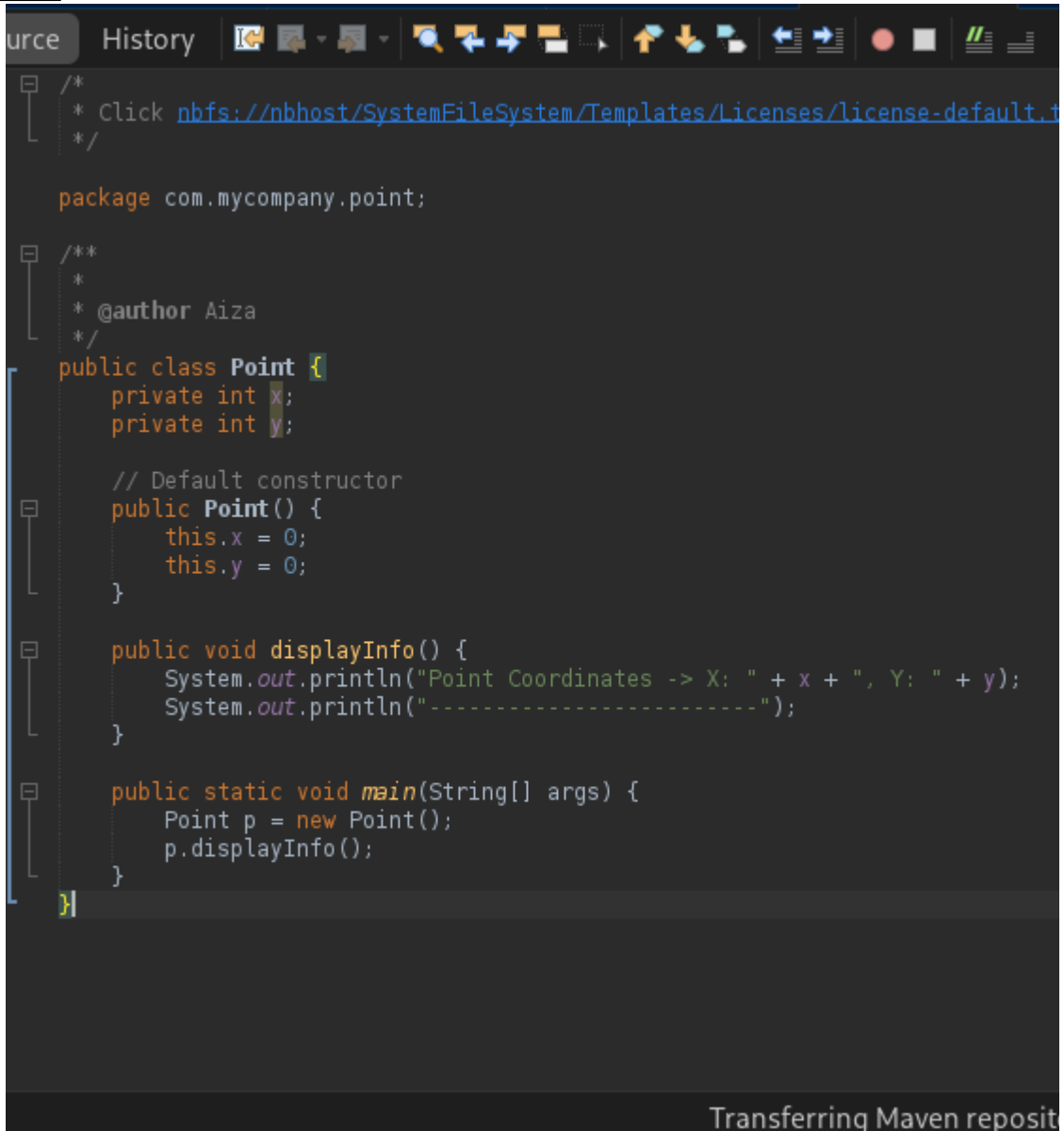
--- exec:3.5.1:exec (default-cli) @ BankAccount ---
Account Number : 0000
Account Holder : Unknown
Balance        : Rs. 0.0
-----
Account Number : HBL-10234
Account Holder : Zainab Hussain
Balance        : Rs. 0.0
-----
Account Number : MCB-50871
Account Holder : Sana Rehman
Balance        : Rs. 75000.5
-----
Account Number : UBL-30912
Account Holder : Maryam Nawaz
Balance        : Rs. 120000.0
-----

BUILD SUCCESS
-----
Total time: 2.076 s
Finished at: 2026-02-22T16:04:57+05:00
-----
```

Output

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Task # 2:



The screenshot shows an IDE window with a dark theme. The top bar includes tabs for 'Source' and 'History', and a toolbar with various icons. The main editor area displays Java code for a class named 'Point'. The code includes a package declaration, a Javadoc comment with an @author tag, a class definition with private fields 'x' and 'y', a default constructor, a 'displayInfo()' method, and a 'main()' method. A file explorer on the left shows a project structure with a 'src' folder. At the bottom right, a status bar indicates 'Transferring Maven reposit'.

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to view the code.
 */

package com.mycompany.point;

/**
 *
 * @author Aiza
 */
public class Point {
    private int x;
    private int y;

    // Default constructor
    public Point() {
        this.x = 0;
        this.y = 0;
    }

    public void displayInfo() {
        System.out.println("Point Coordinates -> X: " + x + ", Y: " + y);
        System.out.println("-----");
    }

    public static void main(String[] args) {
        Point p = new Point();
        p.displayInfo();
    }
}
```

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OutPut:

```
cd /home/rehan/NetBeansProjects/point; JAVA_HOME=/usr/lib/jv
WARNING: A terminally deprecated method in sun.misc.Unsafe h
WARNING: sun.misc.Unsafe::staticFieldBase has been called by
WARNING: Please consider reporting this to the maintainers o
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in
Scanning for projects...

-----< com.mycompany:point >-----
[ ] Building point 1.0-SNAPSHOT
    from pom.xml
    -----[ jar ]-----
[ ] --- resources:3.3.1:resources (default-resources) @ point ---
    skip non existing resourceDirectory /home/rehan/NetBeansProj
[ ] --- compiler:3.13.0:compile (default-compile) @ point ---
    Recompiling the module because of changed source code.
    Compiling 1 source file with javac [debug release 25] to tar
[ ] --- exec:3.5.1:exec (default-cli) @ point ---
    Point(0, 0)
    -----
    BUILD SUCCESS
    -----
    Total time: 2.851 s
    Finished at: 2026-02-22T15:18:14+05:00
    -----
```

Output

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Task # 3



The screenshot shows an IDE window with a dark theme. The top bar includes a 'Source' tab and a 'History' tab. Below the tabs is a toolbar with various icons for file operations and navigation. The main editor area displays Java code for a class named 'Student' in the package 'com.mycompany.student'. The code includes a Javadoc comment with an @author tag for 'Aiza', private fields for 'name', 'rollNumber', and 'cgpa', and three constructors: a default constructor, a parameterized constructor with three arguments, and an overloaded constructor with two arguments where 'cgpa' defaults to 0.0. A status bar at the bottom right indicates 'Transferring Maven reposit'.

```
Source History
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to view the source code.
 */
package com.mycompany.student;

/**
 *
 * @author Aiza
 */
public class Student {
    private String name;
    private int rollNumber;
    private double cgpa;

    // Default constructor
    public Student() {
        this.name = "";
        this.rollNumber = 0;
        this.cgpa = 0.0;
    }

    // Parameterized constructor (3 params)
    public Student(String name, int rollNumber, double cgpa) {
        this.name = name;
        this.rollNumber = rollNumber;
        this.cgpa = cgpa;
    }

    // Overloaded constructor (2 params) - cgpa defaults to 0.0
    public Student(String name, int rollNumber) {
        this.name = name;
        this.rollNumber = rollNumber;
        this.cgpa = 0.0;
    }
}

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```

Output:

```
Building Student 1.0-SNAPSHOT
  from pom.xml
-----[ jar ]-----

--- resources:3.3.1:resources (default-resources) @ Student ---
skip non existing resourceDirectory /home/rehan/NetBeansProj

--- compiler:3.13.0:compile (default-compile) @ Student ---
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to target

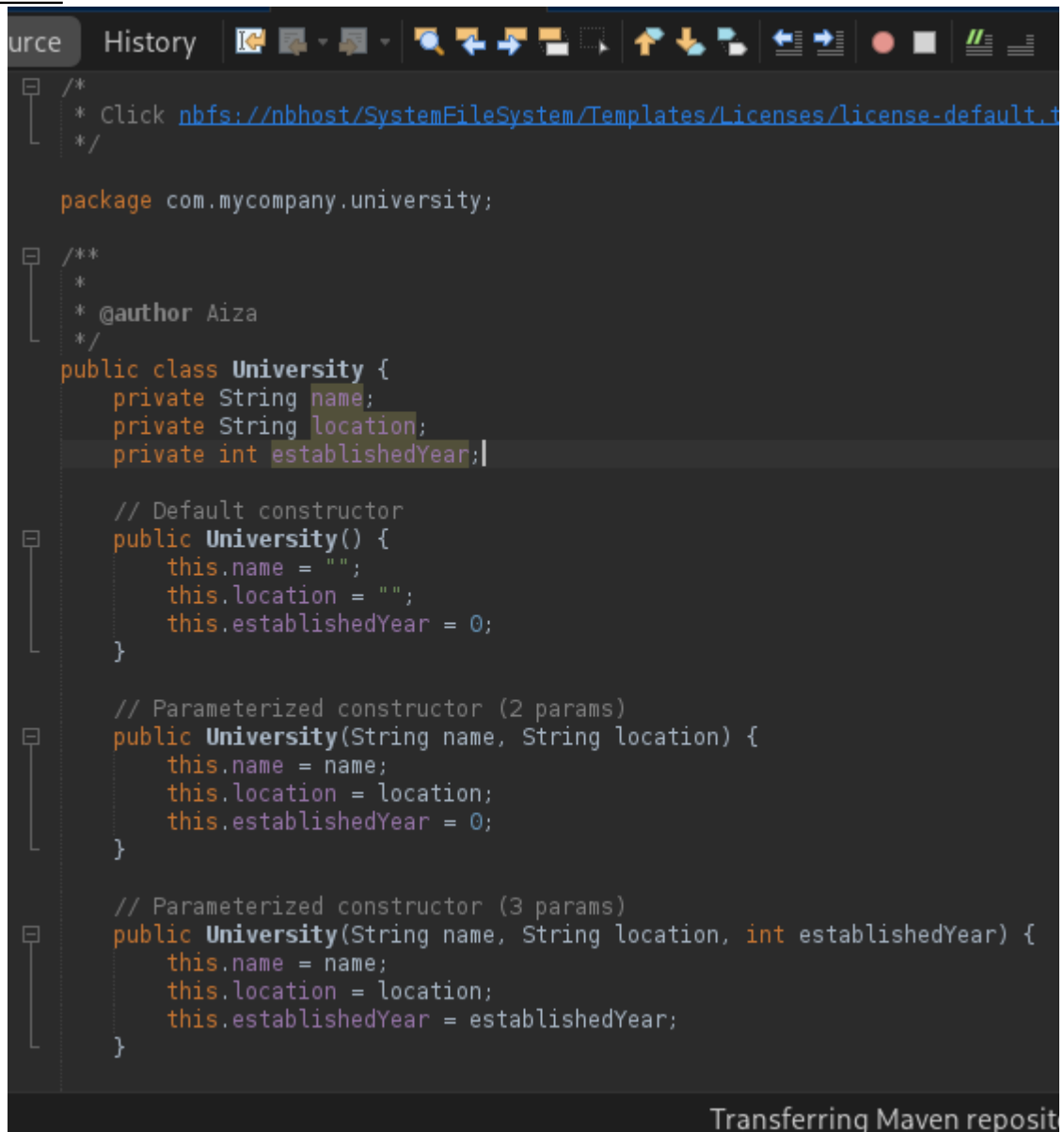
--- exec:3.5.1:exec (default-cli) @ Student ---
Name      :
Roll Number: 0
CGPA      : 0.0
-----
Name      : Ayesha Tariq
Roll Number: 2301
CGPA      : 3.9
-----
Name      : Fatima Malik
Roll Number: 2302
CGPA      : 0.0
-----

BUILD SUCCESS

Total time: 2.608 s
Finished at: 2026-02-22T16:09:15+05:00
-----
```

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Task # 4:



The screenshot shows an IDE window with a dark theme. The top toolbar includes icons for source code, history, and various editing actions. The code is in a file named `license-default.t`. It defines a `University` class with three constructors. The class has three private attributes: `name`, `location`, and `establishedYear`. The first constructor is a default constructor that initializes all attributes to empty strings or zero. The second constructor takes `name` and `location` as parameters. The third constructor takes `name`, `location`, and `establishedYear` as parameters. The IDE status bar at the bottom right indicates 'Transferring Maven reposit'.

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.t
 */

package com.mycompany.university;

/**
 *
 * @author Aiza
 */
public class University {
    private String name;
    private String location;
    private int establishedYear;

    // Default constructor
    public University() {
        this.name = "";
        this.location = "";
        this.establishedYear = 0;
    }

    // Parameterized constructor (2 params)
    public University(String name, String location) {
        this.name = name;
        this.location = location;
        this.establishedYear = 0;
    }

    // Parameterized constructor (3 params)
    public University(String name, String location, int establishedYear) {
        this.name = name;
        this.location = location;
        this.establishedYear = establishedYear;
    }
}
```

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OutPut:

```
Recompiling the module because of changed source code.
- Compiling 1 source file with javac [debug release 25] to tar
[ ] --- exec:3.5.1:exec (default-cli) @ University ---
University Name :
Location       :
Established Year : 0
-----
University Name : University of the Punjab
Location       : Lahore
Established Year : 0
-----
University Name : Quaid-i-Azam University
Location       : Islamabad
Established Year : 1967
-----
University Name : University of Karachi
Location       : Karachi
Established Year : 1951
-----
University Name : NUST
Location       : Islamabad
Established Year : 1991
-----
-----
BUILD SUCCESS
-----
Total time: 1.985 s
Finished at: 2026-02-22T16:07:31+05:00
-----
-----
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```